

Course design.

Basics	Systems	Scaling laws	Data	Algorithms
Tokenization	Kernels	Scaling sequence	evaluation	supervised fine-tuning
Architecture	parallelism	Model complexity	Curriculum	Reinforcement learning
Loss function	Quantization	Loss metrics	Transformers	Synthetic data
Optimizer	Activation checks	parameter form	Filtering	verifications
Learning rate	CPU offload		Deduplication	
	inference		Mixing	

Byte-pair Encoding (BPE) tokenizer

use bytes instead of tokens:
 ⇒ directly token-free approach.

no Models Architecture
 edge byte
 not scaled up yet for to transformers

transformer variables:

* Activation functions: ReLU, SwiGLU

* pos encoding: sinusoidal RoPE

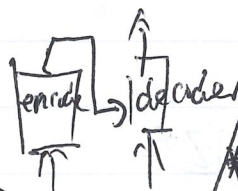
* normalization: layer norm / RMSNorm

* placement of norm: pre-norm / vs / postnorm

* MLP / FNN: dense / sparse / MoE

* Attention: full / self / flash / pytorch / Radix - attention

* low-dim: MLA / GQA / flash: * hybrid transformer: Hyena / Trans-Vnet



Train:

* optimizer: Adam / SWAP

* LR Rate schedule:

* Batch size: critical batch (cosine-ws)

* regulation: dropout / weight decay

* hyperparameter: grid search